Formalizing Events and Situations in Language: A Linguistic Analysis of Event Calculus and Situation Calculus

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1 Introduction

Human language allows speakers to describe actions, events, situations, and change in a flexible and often minimally specified manner. The same sentence can give rise to different interpretations, often due to certain information being left unstated. Human discourse furthermore frequently requires the revision of previously drawn conclusions when new information becomes available. Linguistic expressions encoding phenomena such as aspectual distinctions, causal relations, or temporal sequencing typically rely on shared assumptions and implicit inferences to determine how what is described unfolds over time and affects the state of the world.

A central concern of linguistic semantics is how interpretations of linguistic expressions can be represented in a way that makes their underlying structure explicit and allows for consistent logical reasoning. While human language users are often able to reason about what sentences describe without difficulty, often unconsciously filling in unstated information and drawing implicit conclusions, formal representations of meaning must explicitly encode information that is often left implicit in linguistic expressions. In particular, factors such as temporal structure, causality, and state changes must be made explicit in order to allow for precise reasoning. Different formal systems may therefore impose different assumptions or require different degrees of additional information when used to formalize the same linguistic material, potentially leading to different inferences and different alignments with intuitive interpretations.

This paper investigates how well two formal systems, Event Calculus and Situation Calculus, succeed in accurately representing linguistic expressions about actions, events, situations and change, and comparatively analyzes their primary differences. Following an introduction to the semantic background that motivates the interpretation of sentences as abstract representations, the core methodology of each system is introduced. This allows for a comparative analysis of the assumptions they make, the rules they impose, and their capacity to model linguistic phenomena related to aspect, causality, and temporal structure, including the limitations each system encounters in doing so.

Based on this analysis, this paper draws substantial conclusions; primarily that Event Calculus and Situation Calculus do not merely differ in formal structure, but model fundamentally different stages of discourse interpretation. Event Calculus captures the incremental and revisable nature of interpretation under incomplete information, while Situation Calculus models fully specified and determinate narratives. This paper argues that this core distinction explains their respective strengths and limitations in representing linguistic meaning, and that their apparent differences are best understood as reflecting distinct epistemic perspectives rather than competing ontologies. The paper further claims that Event Calculus's broader expressive range relative to Situation Calculus is partly achieved through representational accommodations within a system whose primitive ontology is fundamentally not designed for expressing the linguistic phenomena examined; despite the calculus having the capability to do so. Finally, this paper also discusses two fundamental limitations shared by both calculi with regard to their ability to accurately map linguistic expressions and typical reasoning processes encoded within language.

2 Semantic Background

2.1 Donald Davidson

Both Event Calculus and Situation Calculus are formulated within first-order predicate logic (FOL), which they use as a formal language in which their axioms and representations are stated. For the purpose of representing linguistic expressions, however, an additional semantic theory is required to motivate how sentences in human language are mapped onto appropriate logical forms expressible in FOL, which can then serve as input to both calculi. This has long been a research goal, with many contributions published on the topic, but Davidson [1967] provides the most relevant theories for such a mapping through his event-based analysis of action sentences in *The Logical Form of Action Sentences*. This section summarizes points from that work that are necessary to understand how one gets from human language sentences to logical representations usable in the calculi.

Davidson critiques prior attempts to reason about events and action sentences by reducing them to claims about agents bringing about resulting states of affairs. Such representations appear in FOL as

$$\text{BroughtAbout}(x, p), \tag{1}$$

where an entity x brings about some state of affairs p as the result of an action A. Davidson argues that this form is inadequate for representing many common expressions, as illustrated by the following examples:

(a) Cass walked to the store.

(b) Smith coughed.

(c) The doctor removed the patient's appendix.

(d) Jones raised his arm.

In (a), representing the sentence as "Cass brought it about that Cass was at the store" fails to encode the manner of the action (walking). In (b), there is no clear resulting state of affairs brought about by coughing. In (c), "The doctor brought it about that the patient had no appendix" could be true even if the doctor merely arranged for the operation to be performed by another doctor. In (d), "Jones brought it about that Jones's arm was raised" may be true even if Jones raised his arm using his other hand, which is not the usual understanding of "raising". In each case, the paraphrase allows for situations in which the original sentence would be false. Davidson concludes that representations of the form in (1) impose truth conditions not aligned with the intuitive understanding of the underlying sentences.

Davidson further notes that multiple sentences may refer to the same underlying event. Consider the following examples:

(i) Amundsen flew to the Morning Star.

(ii) Amundsen flew to the Evening Star.

(iii) I am writing my name on a piece of paper with the intention of writing a cheque with the intention of paying my gambling debt.

Since "Morning Star" and "Evening Star" refer to the same celestial body (Venus), it would be unclear, if simply presented with the statements, whether sentences (i) and (ii) describe one event or two if one attempted to formally model them under prior frameworks. Likewise, sentence (iii) can be paraphrased by statements describing only parts of the action, such as "I am writing a cheque" or "I am paying my debt". If viewed in isolation, it would be unclear that these two statements are both referring to the singular underlying event in (iii).

Davidson's solution is to treat events as entities in their own right that are located within space and time. This is supported by anaphoric reference. Consider the following discourse:

(A) Jones buttered the toast in the bathroom with a knife at midnight.

(B) Please tell me more about it.

Here, "it" refers to the event itself rather than to any participant or nominal phrase from the prior sentence. Davidson therefore proposes that events be represented explicitly in FOL. Thus

$$\text{Kicked}(\text{Shem}, \text{Shaun}) \tag{2}$$

is reformulated as

$$\exists x \left(\text{Kicked}(\text{Shem}, \text{Shaun}, x)\right) \tag{3}$$

where x denotes an event. This also allows for adverbial modification such as

$$\exists x \left(\mathrm{Kicked}(\mathrm{Shem}, \mathrm{Shaun}, x) \land \mathrm{Violent}(x) \land \mathrm{Quick}(x)\right) \tag{4}$$

This conception of events forms the basis for the event entities later used in Event Calculus.

2.2 Further Development: Neo-Davidson

Subsequent work has refined Davidson's proposal by investigating the internal structure of events as encoded in human language. Pustejovsky [2021] provides an overview of these developments in *The Role of Event-Based Representations and Reasoning in Language*, emphasizing that linguistic expressions systematically encode distinctions within event structure that go beyond the mere introduction of an event variable.

A central distinction concerns the temporal type of events. Pustejovsky observes that linguistic expressions encode different kinds of temporal structure. Some events are instantaneous and occur at a discrete time point, while others extend over a period of time. Moreover, events may either culminate in a natural endpoint or lack any inherent termination.

Building on this observation, Pustejovsky adopts and further develops the traditional classification between *telic* and *atelic* eventualities. Telic events are characterized by an inherent endpoint or culmination, as in achievements and accomplishments (e.g., "build a house" or "reach the summit"). Atelic events, by contrast, lack a clear boundary and include processes and states (e.g. "run" or "know French"). This distinction captures a fundamental difference between events that culminate and those that merely persist over an interval. The existence of such aspectual contrasts in language provides a principled justification for treating event-denoting expressions as temporally structured entities, rather than as temporally neutral predicates.

Beyond temporal classification, Pustejovsky emphasizes that event structure is encoded across multiple lexical categories, not only in verbal predicates. Events may be introduced or presupposed by a variety of lexical categories:

(i) Verbal event predicates:
The missile **sank** the ship.

(ii) Nominal event-denoting expressions:
The **meal** was after the speech.

(iii) Nominalizations:
The **explosion** occurred at noon.

(iv) Adjectival predicates implying prior events:
Mary closed the **open** door.

(v) Prepositional or stative predicates describing resultant states:
John is **on board**.

(iv) indicates a "door opening" event having occurred, whereas (v) indicates an event of John boarding having taken place. These discussed distinctions are significant for formal modeling. First, the telic/atelic contrast directly affects how change, culmination, and persistence must be represented in a reasoning system. Event Calculus, for example, processes events differently depending on whether they are construed as culminating or ongoing. Second, the fact that events may be introduced across diverse lexical categories implies that a mapping from linguistic expressions to first-order logic cannot rely solely on surface verbs. A semantic analysis must determine when an event entity is introduced, presupposed, or implied by the structure of the sentence.

Neo-Davidsonian approaches thus extend Davidson's original insight by demonstrating that event variables are not merely logical conveniences but reflect structured semantic information encoded in language. This structured event representation provides the bridge between linguistic interpretation and the formal ontologies presupposed by Event Calculus and Situation Calculus.

The research of both Davidson and Pustejovsky provides more extensive detail about the proper modeling of linguistic expressions onto FOL; however, this provides the theoretical background used in this paper to model language onto FOL to then be processed by Event Calculus and Situation Calculus.

3 Event Calculus

3.1 Overview

The previous section established the semantic background for mapping linguistic expressions describing actions and events onto structured logical representations. This section now introduces the first of the two formal systems examined in this paper: the *Event Calculus*.

The version of Event Calculus adopted here follows the formalism presented in Part 2 of *The proper treatment of events* by van Lambalgen and Hamm [2005], which introduces the core predicates and axioms used to represent events, temporal structure, and change within a first-order framework. Van Lambalgen and Hamm situate Event Calculus within a tradition that emphasizes a computational, rather than purely model-theoretic, notion of meaning, where meaning is tied to inferential consequences and reasoning about change.

Historically, Event Calculus was developed for applications in artificial intelligence, temporal databases, and logic programming, contexts in which explicit reasoning about events and their effects is central.

Although Event Calculus does not originate within linguistic semantics, its ontology includes event entities and time-dependent properties in a manner that is compatible with Davidsonian-style event representations. It therefore provides a formally precise framework within which linguistically motivated event structures can be embedded and evaluated.

For the purpose of mapping linguistic expressions onto both Event Calculus and Situation Calculus in this paper, a Davidsonian or Neo-Davidsonian semantic framework is assumed as an interface layer. Although not explicitly shown, sentences describing actions and events are first translated into first-order representations that introduce event variables in a Davidsonian/Neo-Davidsonian manner. These event-structured representations are then embedded within the ontological and predicate structure of the respective calculus. This methodological choice ensures that the formal representations evaluated in both systems are derived from independently motivated linguistic theory, rather than being tailored to the internal ontological commitments of either calculus.

This section first outlines the ontology and principal predicates of Event Calculus. It then covers the core axioms of the calculus and concludes by discussing the central semantic assumptions that the calculus imposes when formally representing meaning.

3.2 Core Ontology

Event Calculus is formulated in terms of three primary ontological categories that structure its representations: *events*, *time points*, and *fluents*. These entities serve as the arguments of the core predicates through which the calculus produces its models.

3.2.1 Events

Events e are treated as discrete occurrences that take place at specific time points. They are the primary sources of change within the system: an event may initiate or terminate a fluent. In this respect, events function as causal triggers that bring about transitions in state.

3.2.2 Time

Time is typically represented as a set of discrete, linearly ordered time points t. Events are said to occur at discrete times, and the truth of fluents is evaluated relative to these time points. The explicit representation of time allows the calculus to model persistence and change in a formally precise manner.

3.2.3 Fluents

Fluents are time-dependent properties that may hold or fail to hold at a given time. They represent states or conditions of the world that can be affected by events. Unlike events, which occur at specific time points, fluents can extend over intervals and are subject to the principle of inertia: once initiated, a fluent continues to hold until an event explicitly terminates it or begins a transition process. In a given system, each fluent has a set of initiating events $e+$ and terminating events $e-$.

It is important to note that fluents are not simple boolean entities that merely encode whether they hold at a given time. While they are often modeled as holding or not holding at particular moments, and as being initiated and terminated by events, their temporal development need not always reduce to a discrete switch between two fixed states. In some cases, a fluent may undergo a transitional phase extending over an interval, and thus cannot be fully captured in purely binary terms.

3.3 Core Predicates

Event Calculus is formulated using a fixed set of first-order predicates that define the interaction between events, time points, and fluents. The core predicates are listed below.

EC-1 Initially(f)
Fluent f holds initially (i.e., at the initial time of the narrative/model).

EC-2 HoldsAt(f, t)
Fluent f holds at time t.

EC-3 Happens(e, t)
Event e occurs at time point t.

EC-4 Initiates(e, f, t)
Event e initiates fluent f at time t.

EC-5 Terminates(e, f, t)
Event e terminates fluent f at time t.

EC-6 Clipped(t_1, f, t_2)
There exists an event occurring between t_1 and t_2 that terminates fluent f.

EC-7 Declipped(t_1, f, t_2)
There exists an event occurring between t_1 and t_2 that initiates fluent f.

EC-8 Releases(e, f, t)
Event e releases fluent f from inertia at time t.

EC-9 Trajectory(f_1, t, f_2, d)
Given a (typically continuous) development starting from fluent f_1 at t, fluent f_2 holds at $t + d$.

Here, **EC-6** and **EC-7** allow for the representation of initiation and termination events without needing to know the exact moment of occurrence, as long as the interval is known. Furthermore, the predicate **EC-8** allows for the beginning of the previously mentioned non-discrete transition processes, as in sentences such as "the water temperature began to rise." **EC-9** furthermore allows for the modelling of a non-discrete process that culminates in a certain state holding. Here $f1$ should be understood to be a "driving force", whereas $f2$ is the concluding state if that force holds for a certain duration.

3.4 Core Axioms

The behavior of the above mentioned predicates is governed by the following axioms, which are valid in all models utilizing Event Calculus.

EC-A-1 Initially(f) $\rightarrow$ HoldsAt$(f,\ 0)$

EC-A-2 HoldsAt$(f,\ r)$ $\wedge$ $r < t$ $\wedge$ $\neg$Clipped$(r,\ f,\ t)$ $\rightarrow$ HoldsAt$(f,\ t)$

EC-A-3 Happens$(e,\ t)$ $\wedge$ Initiates$(e,\ f,\ t)$ $\wedge$ $t < t'$ $\wedge$ $\neg$Clipped$(t,\ f,\ t')$ $\rightarrow$ HoldsAt$(f,\ t')$

EC-A-4 Happens$(e,\ t)$ $\wedge$ Initiates$(e,\ f_1,\ t)$ $\wedge$ $t < t'$ $\wedge$ $t' = t + d$ $\wedge$ Trajectory$(f_1,\ t,\ f_2,\ d)$ $\wedge$ $\neg$Clipped$(t,\ f_1,\ t')$ $\rightarrow$ HoldsAt$(f_2,\ t')$

EC-A-5 Happens$(e,\ s)$ $\wedge$ $t < s$ $\wedge$ $s < t'$ $\wedge$ (Terminates$(e,\ f,\ s)$ $\vee$ Releases$(e,\ f,\ s)$) $\rightarrow$ Clipped$(t,\ f,\ t')$

EC-A-6 Initially(f) $\wedge$ $\neg$Clipped$(0,\ f,\ t)$ $\rightarrow$ HoldsAt$(f,\ t)$

EC-A-7 Happens$(e,\ t)$ $\wedge$ Terminates$(e,\ f,\ t)$ $\wedge$ $t < t'$ $\wedge$ $\neg$Declipped$(t,\ f,\ t')$ $\rightarrow$ $\neg$HoldsAt$(f,\ t')$

EC-A-8 Happens$(e,\ s)$ $\wedge$ $t < s$ $\wedge$ $s < t'$ $\wedge$ (Initiates$(e,\ f,\ s)$ $\vee$ Releases$(e,\ f,\ s)$) $\rightarrow$ Declipped$(t,\ f,\ t')$

It is important to note that **EC-A-6** reflects a restriction of Event Calculus to the fragment concerned solely with instantaneous change. In the full model, which also includes the trajectory axiom (**EC-A-4**), an equivalent persistence principle is derivable from the interaction of the other axioms.

3.5 Core Semantics

Event Calculus imposes several rules for the semantic interpretation of the events, situations, actions and changes modeled in it. The primary principles are as follows:

3.5.1 Closed-World Reasoning

Event Calculus typically adopts a form of closed-world reasoning with respect to the predicate Happens. Events that are not explicitly stated to occur, and whose occurrence is not required to explain stated consequences, are assumed not to have taken place. In the simple scenario where a *Load* event occurs at time 1 and a *Shoot* event at time 10, and no further events are given, this assumption is formalized as follows:

$$\forall e \forall t \left(\text{Happens}(e, t) \leftrightarrow \big((e = \text{Load} \land t = 1) \lor (e = \text{Shoot} \land t = 10)\big)\right) \tag{5}$$

Under this assumption, no additional events, such as an $Unload$ between $t = 1$ and $t = 10$, are assumed to have occurred. This ensures that reasoning about the consequences of the stated events remains determinate.

3.5.2 Minimal Models

Event Calculus is further associated with minimal-model semantics. If a stated event presupposes that certain prior conditions must have held, but those prior conditions could have arisen in multiple ways, the formalization does not arbitrarily select one complete history. Instead, only what is minimally required to render the stated events possible is assumed.

As a consequence, no additional events are postulated beyond those that are strictly necessary to account for the explicitly stated occurrences and their effects. If a *Loaded* fluent is stated to hold at a time t, it is therefore assumed that between 0 and t, a *Loading* event occurred.

It is worth noting that unless explicitly stated, it is not assumed that a given fluent held initially using the $Initially$ predicate.

Furthermore, if there are multiple minimal models that can lead to the necessary fluents holding at a given time, Event Calculus does not arbitrarily pick one of them. It instead imposes a model where one of the potential minimal models is assumed to hold, but which one it is is left unspecified.

Suppose that, in a model using Event Calculus, the fluent *Loaded* is stated to hold at time

10, but in this model, the *Loading* event is split into $LoadLeft$ and *LoadRight* depending on which hand is used to load the gun. The minimal model would therefore be as follows:

$$\exists t < 10 \left(\text{Happens}(\text{LoadLeft}, t) \ \vee \ \text{Happens}(\text{LoadRight}, t)\right) \tag{6}$$

The problematic nature of minimal models is elaborated in the analysis section; more specifically in subsection 6.6

3.5.3 Non-Monotonic Reasoning

Many other *monotonic* systems of reasoning do not allow for the revision of previously drawn conclusions. This is problematic as, as stated in the introduction, humans in discourse frequently revise previously drawn conclusions if new information arises which contradicts them. Event Calculus allows for this to be modeled by using *non-monoticity.*

Suppose a *Loading* and *Shooting* event occur as previously stated. In alignment with the previous principles, the fluent *Dead* would then be assumed to hold; however, if later new information is added that an $Unloading$ event occurred between the time periods of the *Loading* and *Shooting* events, non-monoticity allows for the correction of the conclusion drawn and the new inference that the fluent *Dead* does not hold.

The foundational axioms and semantic rules of Event Calculus determine the general principles governing persistence, initiation, and termination of fluents. However, they do not specify which particular events initiate or terminate which fluents, nor which events occur in a given narrative. Such information must be supplied through domain-specific axioms and occurrence statements. Together with initial conditions, these constitute a domain description within the calculus.

The core axioms and semantic principles of Event Calculus are shown to solve several "problems" which would arise if one were to simply use FOL in this situation. That is further discussed by Van Lambalgen and Hamm.

4 Situation Calculus

4.1 Overview

In this section, the second calculus examined in this paper is introduced: the *Situation Calculus*. The version adopted here follows the presentation in *The Situation Calculus: A Case for Modal Logic* by Lakemeyer [2010], which provides a systematic account of its ontology and reasoning principles.

The Situation Calculus was originally introduced by McCarthy and later refined by Reiter as a formal framework for reasoning about action and change, as discussed by Lakemeyer [2010]. Unlike Event Calculus, which treats events occurring at specific time points as the primary drivers of change, Situation Calculus organizes its ontology around situations, understood as world states resulting from sequences of actions. Rather than indexing properties to time, it represents change by constructing new situations through the execution of actions. In this framework, world evolution is therefore modeled sequentially: each action transforms a prior situation into a successor situation.

While Situation Calculus does not originate from a Davidsonian theory of events, the action occurrences introduced in its ontology can be interpreted as corresponding to event descriptions derived from linguistic analysis. However, unlike Event Calculus, Situation Calculus does not treat events as independent ontological entities subject to modification; instead, it treats actions as operators that generate new situations.

For the purposes of this paper, linguistically motivated event representations are therefore not discarded, but systematically adapted to the ontological commitments of the calculus. Sentences are first analyzed within a Davidsonian or Neo-Davidsonian framework that introduces event variables and decomposes relevant semantic structure. These event-based representations are then mapped onto action terms in Situation Calculus, where the occurrence of an event is represented as the execution of a corresponding action transforming one situation into another. In this way, the linguistic analysis remains methodologically primary, while the formal calculus provides the structural framework within which its consequences are made explicit.

4.2 Core Ontology

Similar to Event Calculus, Situation Calculus is formulated in terms of four primary ontological categories. These are *objects*, *actions*, *situations*, and $fluents$.

4.2.1 Objects

Unlike Event Calculus, Situation Calculus treats objects as first-class individuals in the domain. Actions take objects as arguments, and fluents describe how the properties and relations of those objects vary across situations.

4.2.2 Actions

Actions in the Situation Calculus are first-order terms that, when performed in a situation, produce a successor situation. They can be thought of as representations of state transitions.

4.2.3 Situations

Situations are first-order terms that denote action histories beginning from constant initial situation $S0$. A new situation is constructed by applying the predicate $do(a, s)$, representing the history that results from performing action a in situation s. These can then be layered to represent the situation resulting from a series of actions starting at $S0$ and ending at the current moment.

4.2.4 Fluents

Unlike in Event Calculus, where fluents are seen as first-order entities f, in Situation Calculus, they are predicates with their arguments being an object and a situation. This is shown as follows:

$$F(x, s) \tag{7}$$

This means "fluent F holds of object x in situation s".

4.3 Core Predicates

The core predicates and function symbols in Situation Calculus are:

SC-1 $do(a, s)$ Successor situation function. Returns the situation that results from performing action a in situation s.

SC-2 $Poss(a, s)$ Action precondition predicate. Action a is executable (possible) in situation s.

This is alongside the initial situation $S0$ as well as the fluent predicates $F(x, s)$.

4.4 Core Axioms

The axioms governing all Situation Calculus systems are the following:

SC-A-1 $\mathrm{do}(a_1, s_1) = \mathrm{do}(a_2, s_2) \rightarrow a_1 = a_2 \wedge s_1 = s_2$

SC-A-2 $\forall Q\,[\,Q(S_0) \wedge \forall s, a\,[\,Q(s) \rightarrow Q(\mathrm{do}(a, s))\,]\,] \rightarrow \forall s\, Q(s)$

SC-A-3 $s \sqsubset \mathrm{do}(a, s') \equiv s \sqsubseteq s'$

SC-A-4 $\neg s \sqsubset S_0$

Here $\sqsubset$ is used to denote when one situation precedes another; that meaning that it can be reachable from that situation via one or more actions. $\sqsubseteq$ is used as a shorthand and means $s \sqsubseteq s' \iff s \sqsubset s' \vee s = s'$. $\equiv$ is used in Situation Calculus to show equivalency.

4.5 Core Semantics

4.5.1 Monotonic Reasoning

In direct contrast with Event Calculus, Situation Calculus operates under monotonic reasoning. This means that there is an assumption that the information from prior situations and actions is necessary and sufficient so that the currently modeled state is complete.

4.5.2 Closed-World Assumption of $S0$

Similar to Event Calculus, a closed-world assumption is applied to the initial state $S0$.

The foundational axioms presented above define the structural properties of all Situation Calculus domains, but do not specify how any particular fluent changes as the result of an action, nor when a given action is executable. This information is supplied through domain-specific axioms: *precondition axioms*, which state the conditions under which an action is possible in a situation, and *successor state axioms*, which provide necessary and sufficient conditions for a fluent to hold after an action has been performed.

Lakemeyer further discusses extensions to Situation Calculus for representing an agent's knowledge through an epistemic accessibility predicate K, as well as both explicit and implicit approaches to modeling time, including branching-time operators. As these extensions are not required for the linguistic analysis in this paper, they will not be further expanded upon.

5 Modeling Scenarios

Following the introduction to both systems, this section uses both to model concrete linguistic expressions. The following scenarios were selected because they illustrate linguistic phenomena that bring out the core differences between the two calculi and therefore allow for further analysis: telicity, gradual processes, and belief revision.

5.1 Simple Telic Event

The sentence *"John locked the door"* describes a simple telic event with a clear resulting state change.

5.1.1 Event Calculus

Domain Axioms

$$\mathit{Initiates}(\mathit{Lock},\ \mathit{Locked},\ t) \tag{8}$$

$$\mathit{Terminates}(\mathit{Lock},\ \mathit{Unlocked},\ t) \tag{9}$$

Narrative

$$\mathit{Initially}(\mathit{Unlocked}) \tag{10}$$

$$\mathit{Happens}(\mathit{Lock},\ t_1) \tag{11}$$

$$t_1 < t_2 \tag{12}$$

5.1.2 Situation Calculus

Initial Situation

$$\neg\, \mathit{Locked}(\mathit{door},\ S_0) \tag{13}$$

Precondition Axiom

$$\mathit{Poss}(\mathit{lock}(x),\ s)\ \equiv\ \neg\, \mathit{Locked}(x,\ s) \tag{14}$$

Successor State Axiom

$$\mathit{Locked}(x,\ \mathit{do}(a,\ s))\ \equiv\ a = \mathit{lock}(x)\ \lor\ \big[\mathit{Locked}(x,\ s)\ \land\ a \neq \mathit{unlock}(x)\big] \tag{15}$$

Resulting Situation

$$s_1 = \mathit{do}(\mathit{lock}(\mathit{door}),\ S_0) \tag{16}$$

$$\mathit{Locked}(\mathit{door},\ s_1) \tag{17}$$

5.2 Gradual Process

The sentence *"The soup heated up and boiled"* describes an atelic process (heating) that culminates in a telic endpoint (boiling).

5.2.1 Event Calculus

Domain Axioms

$$Initiates(TurnOnStove,\ Heating,\ t) \tag{18}$$

$$Releases(TurnOnStove,\ Temperature,\ t) \tag{19}$$

$$Trajectory(Heating,\ t,\ Boiling,\ d) \tag{20}$$

$$Terminates(Boil,\ Heating,\ t) \tag{21}$$

$$Initiates(Boil,\ Boiling,\ t) \tag{22}$$

Narrative

$$Initially(\neg\, Heating) \tag{23}$$

$$Initially(\neg\, Boiling) \tag{24}$$

$$Happens(TurnOnStove,\ t_1) \tag{25}$$

$$t_1 < t_1 + d = t_2 \tag{26}$$

By axiom **EC-A-4**, the *Heating* fluent is initiated at t_1 and released from discrete state changes. The *Trajectory* predicate models the continuous development: if *Heating* persists unclipped for duration d, then *Boiling* holds at $t_2 = t_1 + d$.

5.2.2 Situation Calculus

Initial Situation

$$\neg\, Heating(soup,\ S_0) \tag{27}$$

$$\neg\, Boiling(soup,\ S_0) \tag{28}$$

Precondition Axioms

$$Poss(turnOnStove(x),\ s)\ \equiv\ \neg\, Heating(x,\ s) \tag{29}$$

$$Poss(boil(x),\ s)\ \equiv\ Heating(x,\ s) \tag{30}$$

Successor State Axioms

$$Heating(x,\ do(a,\ s))\ \equiv\ a = turnOnStove(x)\ \vee\ \big[Heating(x,\ s)\ \wedge\ a \neq boil(x)\big] \tag{31}$$

$$Boiling(x,\ do(a,\ s))\ \equiv\ a = boil(x)\ \vee\ Boiling(x,\ s) \tag{32}$$

Resulting Situation

(33) $$s_1 = do(turnOnStove(soup),\ S_0)$$

(34) $$s_2 = do(boil(soup),\ s_1)$$

(35) $$Boiling(soup,\ s_2)$$

5.3 Belief Revision

The discourse *"John loaded the gun and fired. But Mary had unloaded it."* presents a sequence in which an initial conclusion must be revised upon the introduction of new information.

5.3.1 Event Calculus

Domain Axioms

(36) $$Initiates(Load,\ Loaded,\ t)$$

(37) $$Terminates(Unload,\ Loaded,\ t)$$

(38) $$HoldsAt(Loaded,\ t) \rightarrow Initiates(Shoot,\ Dead,\ t)$$

(39) $$HoldsAt(Loaded,\ t) \rightarrow Terminates(Shoot,\ Alive,\ t)$$

Narrative (Stage 1) The first sentence is processed:

(40) $$Initially(Alive)$$

(41) $$Happens(Load,\ t_1)$$

(42) $$Happens(Shoot,\ t_2)$$

(43) $$t_1 < t_2 < t_3$$

Under closed-world reasoning, no other events are assumed. By **EC-A-3**, *Loaded* holds at t_2. The conditional domain axiom then yields $Initiates(Shoot,\ Dead,\ t_2)$, and by **EC-A-3**:

(44) $$HoldsAt(Dead,\ t_3)$$

Narrative (Stage 2) The second sentence introduces new information:

(45) $$Happens(Unload,\ t_u)$$

(46) $$t_1 < t_u < t_2$$

The event *Unload* at t_u terminates *Loaded*. By **EC-A-5**, $\mathit{Clipped}(t_1,\ \mathit{Loaded},\ t_2)$ now holds, so *Loaded* no longer holds at t_2. The conditional domain axiom no longer applies, and by non-monotonic revision:

(47) $$\neg\, \mathit{HoldsAt}(\mathit{Dead},\ t_3)$$

5.3.2 Situation Calculus

Initial Situation

(48) $$\mathit{Alive}(\mathit{target},\ S_0)$$

(49) $$\neg\, \mathit{Loaded}(\mathit{gun},\ S_0)$$

(50) $$\neg\, \mathit{Dead}(\mathit{target},\ S_0)$$

Precondition Axioms

(51) $$\mathit{Poss}(\mathit{load}(x),\ s)\ \equiv\ \neg\, \mathit{Loaded}(x,\ s)$$

(52) $$\mathit{Poss}(\mathit{unload}(x),\ s)\ \equiv\ \mathit{Loaded}(x,\ s)$$

(53) $$\mathit{Poss}(\mathit{shoot}(x),\ s)\ \equiv\ \mathit{Loaded}(x,\ s)$$

Successor State Axioms

(54) $$\mathit{Loaded}(x,\ do(a,\ s))\ \equiv\ a = \mathit{load}(x)\ \lor\ [\mathit{Loaded}(x,\ s)\ \land\ a \neq \mathit{unload}(x)]$$

(55) $$\mathit{Dead}(y,\ do(a,\ s))\ \equiv\ [a = \mathit{shoot}(x)\ \land\ \mathit{Loaded}(x,\ s)\ \lor\ \mathit{Dead}(y,\ s)]$$

Resulting Situation The complete action history must be specified from the outset. No intermediate conclusion is drawn and subsequently revised; instead, the unloading is incorporated directly into the situation term:

(56) $$s_1 = do(\mathit{load}(\mathit{gun}),\ S_0)$$

(57) $$s_2 = do(\mathit{unload}(\mathit{gun}),\ s_1)$$

(58) $$s_3 = do(\mathit{shoot}(\mathit{gun}),\ s_2)$$

At s_2, the successor state axiom for *Loaded* yields $\neg\, \mathit{Loaded}(\mathit{gun},\ s_2)$. At s_3, since *Loaded* does not hold in s_2, the precondition $\mathit{Poss}(\mathit{shoot}(\mathit{gun}),\ s_2)$ fails. Even if the shoot action is permitted, the successor state axiom for *Dead* requires $\mathit{Loaded}(\mathit{gun},\ s_2)$, which does not hold. Therefore:

$$\neg \, Dead(target, \; s_3) \tag{59}$$

Both calculi reach the same final conclusion, but through fundamentally different processes. Event Calculus first derives *Dead* and then retracts it upon receiving new information. Situation Calculus never derives *Dead* at all, as the complete history must be known before any reasoning begins. It should therefore be noted that this comparison does not show Situation Calculus to be deficient in modeling the described scenario, but rather that the two systems are modeling different communicative situations. Event Calculus models the discourse as it is incrementally presented to a comprehender, where the first sentence is processed before the second sentence introduces new information. Situation Calculus, by contrast, models the final state of affairs once all information has been received. The difference is therefore not one of correctness but of alignment to different stages of discourse comprehension. This difference is further elaborated in subsection 6.5 of the analysis.

6 Analysis

6.1 Overview

Following the introduction to both systems and the modeling of linguistic expressions, this section provides an analysis of their differences as well as alignment to the intuitive meaning of linguistic expressions.

6.2 Modeling of Time

One of the most substantial points of contrast between the two calculi is how they model temporality. Both systems require that events be ordered relative to one another or, in the case of Event Calculus, be shown as simultaneous. Event Calculus uses ordering constraints on time-point variables, and Situation Calculus employs sequential nesting of actions within situation terms. These requirements represent a form of divergence from human language, where the temporal ordering of events is not always specified. A speaker may say "John traveled a lot and also wrote a novel" without committing to any ordering between these events, yet both calculi would require one to be established in order to model the discourse. Event Calculus; while allowing either a clear order or simultaneous events, lacks a clear method of modeling overlapping events, where they occur partially simultaneously, partially consecutively. In such a case, the underlying events would need to be split into multiple subevents to model when they overlap and when they are occurring in separate timepoints.

Beyond these limitations, however, the two systems diverge in an important respect. Because Event Calculus models time as an independent dimension within which events are located, it is able to represent the passage of time even in the absence of any event occurring. Situation Calculus, by contrast, constructs temporal order solely through the execution of actions. Since situations advance only via the application of the do function, time in Situation Calculus is inseparable from action. A phrase such as "ten minutes passed and nothing happened" can be modeled straightforwardly in Event Calculus, as time points advance while fluents persist through inertia, but lacks a natural representation in Situation Calculus, where no situation term can be constructed without an associated action. This constitutes a notable limitation of Situation Calculus for the representation of linguistic expressions in which temporality is expressed independently of any event or action taking place.

6.3 Aspectual Representation

As shown in the second modeled example, Event Calculus provides a mechanism for representing imperfective aspect. It does so by modeling imperfective events as fluents that extend over a period of time and must themselves be initiated and terminated through the respective predicates. These fluents can then be understood as driving changes via the *Trajectory* predicate as shown in (20), which models the continuous development toward a culmination point. While this mechanism captures certain important properties of imperfective aspect, such as the temporal extension of a process and its interruptibility prior to completion, it does not fully reflect the way imperfective aspect functions in human language. In linguistic expressions, imperfective marking presents an event from an internal perspective, situating the speaker within the ongoing process rather than viewing it from outside as a bounded whole.

Event Calculus does not encode this perspectival distinction; rather, its treatment of imperfective aspect appears to be an accommodation within a system whose event entities are fundamentally perfective in nature, rather than a native implementation of the imperfective as an independent aspectual category. This asymmetry becomes clear when considering what fluents typically represent. In Event Calculus, fluents naturally correspond to static states, those being properties that hold of the world at a given time, such as *Loaded* or *Alive*. When mapping linguistic expressions onto logical form, this aligns most naturally with adjectival predicates, which similarly encode static properties rather than ongoing processes. Imperfective representations, however, are fundamentally processes, not static states. They encode an event as ongoing rather than as a state that simply holds. Representing *Heating* as a fluent alongside *Loaded* therefore conflates two categorically distinct phenomena under the same representational type. The fact that a separate $Trajectory$ predicate must then be introduced to allow for continuous development that imperfective aspect encodes within language directly reveals that the fluent representation is doing work it was not designed for. Perfective events, by contrast, map straightforwardly onto event entities e, which are the calculus's native mechanism for representing bounded occurrences. The asymmetry in representational fit; events for perfectives, fluents forcibly used for imperfectives suggests that Event Calculus is fundamentally a perfective system that accommodates imperfective aspect through secondary workarounds rather than encoding it as an independent category.

This solution in Event Calculus furthermore requires the insertion of a distinct "Start" and "End" events in Event Calculus for imperfective events that function as the initiating and terminating events for the imperfective fluents. This does not entirely reflect human language intuition. Research shows that events conveyed with imperfective aspect are significantly less likely to be perceived as bounded by comprehenders, suggesting that the discrete initiation and termination points required by Event Calculus do not reflect how such events are cognitively represented. Feller et al. [2019].

Situation Calculus, on the other hand, does not in the fundamental form discussed in this paper possess the capacity to represent imperfective aspect at all. Because change is modeled exclusively through discrete actions that yield clearly delineated successor situations, only perfective eventualities that culminate in a definite result can be naturally represented. Atelic eventualities, which by definition lack a natural endpoint, resist this form of representation. This constitutes a significant divergence from human language, in which the imperfective is a fundamental and pervasive aspectual category, but it follows directly from the ontological commitments of the calculus itself.

6.4 Causation

A further point of comparison concerns the representation of causation. Human language does not only describe temporally ordered events; it frequently encodes relations of cause, enablement, and consequence. Sentences such as "John killed Mary", "The push caused her to fall", or "The key opened the door" convey interdependencies between events and resulting states. There is also a fundamental distinction between direct causation, that being when an event directly brings about another event or a resulting state, and enablement, that being when an event or state is necessary so that another event can bring about a resulting event or state. A formal system intended to model linguistic meaning must therefore account for how these aforementioned interdependencies as well as the distinction between direct causation and enablement are structured.

In Event Calculus, causation is represented indirectly through the interaction of events and fluents. An event initiates or terminates a fluent, and the persistence axioms ensure that this change propagates forward in time. Direct causation is represented simply with the $\rightarrow$ connector. Enablement is represented by adding an additional required condition before the implication using the $\wedge$ connector and typically the $HoldsAt$ predicate to express that a certain fluent must hold in order for the causation to occur.

This has the advantage of naturally representing indirect and delayed causation, since fluents persist until terminated and a causal chain may therefore unfold across multiple time points.

Situation Calculus, by contrast, embeds causation within successor state axioms, where each action is associated with necessary and sufficient conditions for a fluent to hold in the resulting situation. Causation is thus tightly localized to discrete state transitions, providing a high degree of determinacy at the cost of requiring indirect or extended causal processes to be decomposed into sequences of discrete actions.

Both systems differ in their methods of representing the distinction between direct causation and enablement. Human language systematically differentiates between actively bringing about a result, as in "John broke the window," and merely creating conditions under which something may occur, as in "John left the window open and the storm broke it." Situation Calculus provides a relatively natural structural account of this distinction through the separation of precondition and successor state axioms. Consider the following formalization. The action $\mathit{leaveOpen}(\mathit{window})$ establishes the precondition for a susequent action.

(60) $$\mathit{Poss}(\mathit{shatter}(\mathit{window}),\ s) \equiv \mathit{Open}(\mathit{window},\ s)$$

(61) $$\mathit{Shattered}(\mathit{window},\ \mathit{do}(a,\ s)) \equiv a = \mathit{shatter}(\mathit{window})\ \vee\ \mathit{Shattered}(\mathit{window},\ s)$$

Here, John's enabling action satisfies the Poss condition without itself producing the result state. The storm's action is what triggers the successor state axiom and produces $\mathit{Shattered}$. The structural distinction between enablement and direct causation is thus encoded in the separation of the two axiom types, and the $Poss$ predicate directly encodes a native representation of enablement.

Event Calculus, by contrast, lacks an analogous structural distinction. To model the same scenario, one must represent the enabling event as initiating a fluent Open that then conditions a subsequent event's initiating effect via $HoldsAt$:

(62) $\quad Initiates(leaveOpen,\ Open,\ t)$

(63) $\quad HoldsAt(Open,\ t) \rightarrow Initiates(shatter,\ Shattered,\ t)$

While this achieves the correct inferential result, it does not formally distinguish the enabling relationship from any other causal sequence in the system. The conditioned *Initiates* clause is structurally identical to how any precondition-dependent causation would be represented. This diverges from human language, where enablement is typically marked explicitly through modal expressions such as "can" or "allow," signaling a categorically different relationship from direct causation. While Event Calculus can as described above use enabling fluents to achieve the same result, it is, similar to the representation of aspect, a workaround allowing for the distinction of enablement in a system that fundamentally does not account for it. Situation Calculus, on the other hand, has a native, intuitive representation of enablement through the *Poss* predicate.

A related limitation shared by both calculi concerns the lexical encoding of causal structure. Language systematically distinguishes causative and inchoative uses of the same verb: "John broke the window" encodes an external agent causing a result, whereas "the window broke" describes the same result state arising without explicit reference to an external cause. This causative/inchoative alternation, discussed extensively in Levin [1993], reflects a semantic distinction between agentive causation and spontaneous change that is encoded in the argument structure of verbs rather than in the logical form of event sequences. Neither calculus has resources to represent this distinction. In Event Calculus, the *Initiates* predicate does not encode whether the initiating event is agentive or spontaneous. Both "John broke the window" and "the window broke" would be represented identically as $Initiates(break,\ Broken,\ t)$. Situation Calculus is equally insensitive. The successor state axiom tracks only whether $Broken(window,\ s)$ holds after $do(break(window),\ s)$, with no structural difference between an agent-caused and a spontaneous transition. This constitutes a systematic gap in the expressive capacity of both systems with respect to causal structure in human language.

6.5 Requirement of Complete Information

A large point of contrast between the two calculi concerns their treatment of the informational status of the past. Situation Calculus represents the current state of the world as the result of a fully specified sequence of actions stemming from the initial situation $S0$. As an example:

(64) $$s_{\text{now}} = do(putonfloor(B),\ do(pickup(B),\ S_0))$$

Examples such as (56-58) are simply a structured way of showing state transitions. (60) shows the object B being picked up and then put on the floor. Critically, it is shown that successor situations all stem back to the neutral "start situation" $S0$ and therefore that the information provided about the course of events is complete. This makes representations such as the third example impossible.

Event Calculus does not require such an explicit enumeration of the entire course of events. Conclusions are drawn based upon what is known, even if that information is not complete at a given moment. Consequently, conclusions drawn about the current state may be revised if new information is introduced, even if that information is with regard to events happening at a prior moment. In this respect, Event Calculus reflects the incremental and revisable character of human discourse more closely than Situation Calculus, which models a world whose action history is assumed to be procedurally complete. It is worth noting that extensions of Situation Calculus such as those briefly mentioned in section 4.5.2 do exist that allow for modeling some of the phenomena unable to be modeled in the base version analyzed in this paper; however, the limitations of the base version do still persist.

In contexts where a narrative is presented as fully specified and determinate, Situation Calculus, even in its base form, may in fact provide a more accurate formal reflection of the linguistic representation than Event Calculus. Because the closed-world assumption applies only to the initial situation $S0$, and because successor state axioms provide necessary and sufficient conditions for each fluent after any action, the truth value of every property at every reachable situation is fully determined by the theory's deductive structure rather than by extra-logical assumptions about minimality. The calculus encodes precisely the sequence of actions explicitly stated, and the completeness of information at each subsequent situation follows as a logical consequence of the biconditional form of the successor state axioms, which simultaneously capture both the effects and non-effects of actions.

Event Calculus, by contrast, relies on minimal-model semantics discussed in 3.5.2 and the circumscription of event occurrences to achieve an analogous completeness, even when no revision occurs, the theory presupposes that only those events strictly required by the stated information are assumed to have taken place: a default-based closure over the entire narrative timeline. In finalized narratives, however, the language does not necessarily encode such epistemic minimality; the speaker's presentation of a complete sequence of events is better reflected by a system in which completeness is structurally guaranteed rather than imposed through defeasible assumptions. The monotonic structure of Situation Calculus therefore avoids introducing interpretive commitments that are not linguistically expressed in the narrative itself. This makes Situation Calculus both preferable for fully complete determinate narratives as well as more broadly to scenarios of low complexity, where the full sequence of relevant actions can be straightforwardly enumerated; in such cases, Situation Calculus avoids the imposition of a minimality requirement that, as discussed in Section 6.6, constitutes a form of inferential bias not reflected in the linguistic material.

6.6 The Problem of Minimal Models

Further expanding on the point in 6.5, Event Calculus attempts to solve for potential incompleteness of prior information through the use of minimal models. This type of modeling, while efficient and providing reasoning capabilities not present in Situation Calculus, fails to accurately reflect the reasoning that occurs in humans while using language. Suppose a model where someone states that they went to a concert, and to allow for this, it is modeled that they must have transported themselves to the concert. Furthermore, there are two possible known methods of transportation: using a helicopter and using public transit. Event Calculus would model the transportation to the concert as follows:

(65) $$\exists t < \mathit{now}\ \big(\mathrm{Happens}(\mathrm{UsePublicTransit}, t)\ \vee\ \mathrm{Happens}(\mathrm{FlyHelicopter}, t)\big)$$

The minimal model principle does not commit to either possibility unless additional, concrete information is provided. From a formal perspective, this is desirable: both possibilities are compatible with the available information. However, psycholinguistic research suggests that human language users do not typically remain agnostic between all logically possible alternatives. Instead, they actively construct inferences guided by plausibility, world knowledge, and probabilistic expectations. During discourse comprehension, readers generate bridging and elaborative inferences that favor the most coherent and contextually plausible interpretation rather than maintaining agnosticism amongst theoretically possible alternatives if none are explicitly shown to be true [Graesser et al., 1994].

In the present example, flying a helicopter to a concert would normally be treated as highly implausible unless specific contextual cues support such an interpretation. Human reasoners therefore tend to privilege the most typical or contextually likely transportation method, even if another one is theoretically possible, rather than maintaining a state of non-commitment to any possible option. Event Calculus, by contrast, preserves all logically compatible histories consistent with minimal assumptions. This reveals a divergence between the calculus' non-monotonic minimality principle and the plausibility-driven inferential processes characteristic of human language understanding. This suggests that minimal-model semantics, while formally elegant, fail to capture a central property of human interpretation: the tendency to privilege plausible over merely possible inferences.

It is worth noting that Situation Calculus does not have the capability to reflect this sense of human cognition and reasoning either, as it cannot model the uncertainty of the transportation method in the first place. Such modeling requires a framework that encodes the plausibility or probability of events and actions when there are multiple potential explanations for the origin of a scenario. Both of these systems are also not designed to reflect human cognition, but rather to enable consistent logical reasoning; however, due to humans being the primary users of language, it is worth noting the divergence between the logical reasoning mechanisms unconsciously employed by humans and those used in formal systems while processing the same linguistic input.

7 Conclusion

The analysis in this paper has shown that Event Calculus and Situation Calculus are not competing formalisms but rather systems suited to fundamentally different conceptions of discourse. The key distinction is not merely technical but reflects a deeper divergence in what each system assumes about the relationship between a speaker, the information available to them, and the linguistic expressions they produce. Event Calculus presupposes that discourse unfolds incrementally and that the information available at any given point may be incomplete or subject to revision. Situation Calculus presupposes that the relevant history of actions is fully specified and that reasoning proceeds over a complete, determinate model. The choice between the two systems therefore depends not on which is formally superior in the abstract, but on which conception of discourse is appropriate to the linguistic material being modeled.

More precisely, the difference between Event Calculus and Situation Calculus is not best understood as a disagreement about the structure of events in the world, but as a difference in the informational perspective they assume. Event Calculus models interpretation under incomplete and revisable information, while Situation Calculus presupposes a fully specified and determinate history. Their divergence is therefore epistemic rather than ontological as they do not compete as alternative descriptions of the same reality, but as models of different stages of discourse interpretation.

Where the linguistic context involves live discourse, partial information, or expressions that require the encoding of temporal passage independently of any occurrence, Event

Calculus provides mechanisms that have no structural equivalent in Situation Calculus. Its treatment of time as an independent dimension and its capacity for non-monotonic revision align more closely with the incremental, revisable character of spoken and unfolding discourse. Event Calculus furthermore extends to a wider range of linguistic phenomena than Situation Calculus, including imperfective aspect and enablement relations; however, as the analysis in sections 6.3 and 6.4 demonstrate, several of these extensions are accommodations within a system whose primitive ontology is fundamentally perfective and action-neutral, rather than native implementations of the phenomena they represent. Event Calculus therefore achieves inferential adequacy across a broader domain than Situation Calculus, without actually using the semantic distinctions that the linguistic material itself encodes.

Situation Calculus, by contrast, is better suited to fully specified narratives. Its monotonic structure and biconditional successor state axioms avoid imposing interpretive commitments, such as the minimality requirement, that are not present in the linguistic material itself. Furthermore, the structural separation of precondition and successor state axioms provides a natural formal analogue to the linguistic distinction between enablement and direct causation, a distinction that Event Calculus can only approximate through intermediate fluent chains.

However, neither system adequately captures one of the most pervasive features of human language comprehension: the role of plausibility in guiding inference. As demonstrated by the minimal-model analysis in section 6.6, Event Calculus maintains formal agnosticism among all logically compatible explanations for unstated prior events, whereas psycholinguistic evidence consistently shows that human comprehenders privilege contextually plausible interpretations over merely possible ones [Graesser et al., 1994]. Because language comprehension routinely relies on plausibility-based inferences, a formal reasoning system should account not only for logical compatibility but also, where possible, reflect the cognitive processes underlying human inference if the goal is to accurately represent linguistic meaning. This is an area in which Event Calculus could be further developed.

Situation Calculus does not improve upon this limitation, as it cannot represent the uncertainty of outcome at all. This shared gap points toward a broader conclusion: a formally complete account of how linguistic meaning is constructed during comprehension would require mechanisms that encode not only logical compatibility but also the probabilistic and contextual factors that guide human inference.

Furthermore, as discussed in section 6.4, neither calculus fundamentally accounts for the difference between causative and inchoative verbal phrases. This leads to both calculi lacking the ability to distinguish between when a resulting state occurs due to an agent actively bringing about it or due to a spontaneous happening, despite that being fundamental to the nature of language.

The contribution of this paper has therefore been fourfold: to demonstrate that both calculi are viable tools for the formal representation of linguistic expressions; to show that their applicability is conditioned by the type of discourse being modeled, reflecting an epistemic rather than ontological difference, to argue that Event Calculus's broader expressive range is partly achieved through representational accommodations that sacrifice the semantic distinctions motivating them, and to point out fundamental linguistic phenomena not accounted for in either calculus. Taken together, these findings suggest that neither system's primitive ontology is natively suited to the full range of phenomena encoded in human language, and that within the realm of linguistic expressions that can be represented within the calculi, neither is inherently superior. They simply differ, and the more suited calculus should therefore be chosen based on the specific linguistic phenomena being modeled.

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